

JAIMIN BABARIA

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Education

University of Illinois Chicago

Master of Science in Computer Science GPA: 4.0/4.0

Aug 2024 – May 2026

Chicago, Illinois

Nirma University

Bachelor of Technology in Computer Science and Engineering

Aug 2021 – May 2024

Ahmedabad, India

Experience

UI Health – University of Illinois Hospital

Data & Operations Assistant

Aug 2024 – May 2026

Chicago, IL

- Cut manual data-entry time by 50% and improved record accuracy by building Python and advanced Excel macro/VBA automations that ingested, validated, and reconciled data in the EP Evaluator system.
- Designed data-driven inventory reporting for medical-supply tracking, surfacing live stock levels and reorder thresholds that kept procurement on schedule across multiple hospital departments.
- Supported patient-data management and daily operations in the Epic EHR, maintaining HIPAA-aligned handling standards across administrative workflows.

TatvaSoft

Software Engineer Intern

Jan 2024 – Jun 2024

Ahmedabad, India

- Profiled and optimized REST APIs and data-processing pipelines for a portfolio-management platform, cutting average response time 30% and improving throughput 20% via query optimization, caching, and endpoint restructuring.
- Integrated Python services with Django and SQL databases in an Agile environment, reducing the production bug rate 15% and accelerating data-feature deployment cycles.

Nirma University

Team Lead – Margdarshak: IoT-driven Safe Route Navigation (State-Funded)

Aug 2023 – Dec 2023

Ahmedabad, India

- Led a 4-engineer team building a state-funded IoT navigation system, programming ESP32 and SIM800L modules over Arduino to stream JSON sensor telemetry (rain, humidity, temperature) into an AWS storage pipeline for real-time route visualization.
- Improved data accessibility 40% by designing a CI/CD-ready configuration interface for live sensor requests, lowering response latency across distributed IoT nodes.

Projects

GitHub PR Automated Code Reviewer | Python, FastAPI, AWS, GitHub API, OpenAI

- Built an event-driven AWS pipeline (SNS, SQS with DLQ, Lambda, DynamoDB, CloudWatch) that automatically reviews GitHub pull requests with the OpenAI API and posts inline code-review comments via the GitHub Apps API, removing manual first-pass review effort.
- Exposed review history, per-repository stats, and aggregate metrics through a FastAPI REST API backed by DynamoDB with status tracking across the full job lifecycle, enabling end-to-end auditability of automated reviews.

CTA Transit Equity & Reliability Pipeline | Python, PostgreSQL/PostGIS, dbt, Airflow, Docker, Tableau

- Built a transit-reliability pipeline polling the CTA Bus Tracker API every 60s and joining GTFS schedule data in PostGIS, with dbt-modeled fact tables computing on-time performance, headway adherence, and ghost-bus rates per route via time-series window functions, orchestrated as a Dockerized Airflow DAG.
- Quantified reliability disparities across 77 Chicago community areas by overlaying ACS demographic data on CTA arrival events, surfacing LOD-driven reliability KPIs through a published Tableau dashboard with Great Expectations checkpoints gating mart publication.

Auditing Socioeconomic Bias in LLM-Based Financial Advice | Python, Fairlearn, scikit-learn

- Audited GPT-4o-mini, LLaMA-3, and Gemini Flash for socioeconomic bias in financial-advice generation, quantifying response disparities with Demographic Parity, Disparate Impact Ratio, and Equalized Odds metrics through Fairlearn.
- Reduced output disparity by assembling a controlled prompt suite encoding SES signals via zip code, occupation, and education, then applying fairness prompting, an SES logistic-regression classifier, and post-processing corrections.

Technical Skills

Languages: Python, Java, SQL, C++, JavaScript

Data Engineering: PostgreSQL, PostGIS, dbt, Apache Airflow, Pandas, NumPy, ETL Pipelines, Tableau, Power BI

Cloud & DevOps: AWS (Lambda, SQS, SNS, DynamoDB, S3, EC2, CloudWatch), Docker, Git, GitHub

Frameworks & Databases: FastAPI, Django, Flask, React.js, Node.js, REST APIs, MySQL, MongoDB

Machine Learning: Scikit-learn, PyTorch, TensorFlow, Hugging Face, Fairlearn, RAG